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System and method for generating a virtual experience in a virtual environment

Abstract

A system includes a memory and a processor coupled to the memory. The processor receives a first user credential associated with a first user and authorizes an avatar of the first user to enter a virtual environment. The processor receives a first selection of an experience theme and a second selection of at least one product type from the selected experience theme. The processor determines a first data file that provides a first benefit count to the first user when a first amount of data objects is transferred from a first data file of the first user to a second data file. The processor indicates to the first user that the first data file provides the first benefit count to the first user, receives a command to obtain the first data file, and initiates a registration process to register the first user to obtain the first data file.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7396281	12/2007	Mendelsohn	725/23	G06Q 30/0209
8082297	12/2010	Syvain et al.	N/A	N/A
8424075	12/2012	Walsh	726/14	H04L 63/0263
8433656	12/2012	Evans	705/902	G06F 21/6236
8631417	12/2013	Mendes Da Costa et al.	N/A	N/A
9165426	12/2014	Jones et al.	N/A	N/A
9218468	12/2014	Rappaport	N/A	H04L 63/12
9342211	12/2015	Deluca et al.	N/A	N/A
9588783	12/2016	Syvain	N/A	G06F 9/453
10063916	12/2017	Hardy	N/A	H04N 13/332
10326667	12/2018	Jones et al.	N/A	N/A
10587584	12/2019	Vikramaratne et al.	N/A	N/A
10610787	12/2019	Hamilton, II et al.	N/A	N/A
10623446	12/2019	Stoler	N/A	H04L 63/101
10664903	12/2019	Haitani	N/A	G02B 27/017
10706155	12/2019	Veselov	N/A	G06F 9/45558
10742649	12/2019	Hook, Jr.	N/A	H04L 63/0272
10831805	12/2019	Buckingham	N/A	G06F 21/31
10931650	12/2020	Mccown et al.	N/A	N/A
10981069	12/2020	Kawachiya et al.	N/A	N/A
10986087	12/2020	Watson	N/A	G06F 3/017
11043219	12/2020	Walters	N/A	G10L 21/013
11170003	12/2020	Wu et al.	N/A	N/A
11227060	12/2021	John	N/A	G06F 21/606
11282066	12/2021	Spender	N/A	G06Q 20/3276
11429939	12/2021	Borunda	N/A	G06Q 20/10
11546322	12/2022	Hardjono	N/A	H04L 67/131
11568614	12/2022	Albero	N/A	G06F 3/011
11769198	12/2022	Doss	705/35	G06F 16/739
11841946	12/2022	Durvasula	N/A	G06F 21/53
11930005	12/2023	Young	N/A	H04L 63/1483
11954030	12/2023	Singh	N/A	G06F 3/067
11995620	12/2023	Borunda	N/A	H04L 63/083

12028348	12/2023	Varma	N/A	G06T 11/00
12063238	12/2023	Albero	N/A	G06F 18/214
12093355	12/2023	Bhatia	N/A	G06F 21/31
12093410	12/2023	Chavez	N/A	H04N 7/188
12106431	12/2023	Padiyar	N/A	G06V 10/141
12112301	12/2023	Padiyar	N/A	G06Q 20/4014
12112438	12/2023	Gupta	N/A	G06F 21/64
12113789	12/2023	Patel	N/A	G06T 11/00
12126606	12/2023	Ingram	N/A	H04L 63/08
12149624	12/2023	Akkapeddi	N/A	H04L 63/102
12190298	12/2024	Moore	N/A	G06Q 20/389
12231462	12/2024	Merdassa	N/A	H04L 63/1433
12244605	12/2024	Patel	N/A	H04L 63/107
12251638	12/2024	Horstman	N/A	A63F 13/79
12259954	12/2024	Dave	N/A	G06T 13/40
12262201	12/2024	Ganesan	N/A	H04W 12/068
12273327	12/2024	Race	N/A	H04L 67/131
12273333	12/2024	Agrawal	N/A	H04L 63/08
2007/0254631	12/2006	Spooner	N/A	N/A
2009/0234948	12/2008	Garbow	709/225	A63F 13/358
2010/0029371	12/2009	Medvinsky	463/25	H04L 67/51
2010/0037147	12/2009	Champion	715/751	G06F 21/316
2010/0057715	12/2009	Bates	707/E17.014	G06F 21/31
2010/0138755	12/2009	Kulkarni	715/757	G06F 9/455
2010/0250383	12/2009	Frazier et al.	N/A	N/A
2010/0299747	12/2009	Hamilton, II	726/19	H04L 9/3247
2011/0010675	12/2010	Hamilton, II	715/850	A63F 13/335
2011/0072367	12/2010	Bauer	N/A	N/A
2011/0165939	12/2010	Borst	463/29	A63F 13/63
2013/0014033	12/2012	Hamick	715/757	G06Q 50/01
2013/0156399	12/2012	Joy	386/E9.011	H04N 21/8545
2013/0339906	12/2012	Barthelt	715/850	G06Q 30/0643
2014/0026078	12/2013	Dawson et al.	N/A	N/A
2014/0096036	12/2013	Mohler	715/753	G06Q 10/10
2014/0176534	12/2013	Vignocchi	345/419	A63F 13/21
2014/0181706	12/2013	Vignocchi	715/764	A63F 13/79
2014/0282874	12/2013	Bennette	726/3	H04L 63/102
2015/0143487	12/2014	Nathan	726/6	H04L 63/08
2016/0055758	12/2015	Francis	434/236	G09B 7/00
2016/0269508	12/2015	Sharma	N/A	H04L 63/0884
2016/0350609	12/2015	Mason	N/A	G06T 15/20
2017/0087473	12/2016	Siegel	N/A	G07F 17/3293
2017/0103571	12/2016	Beaurepaire	N/A	G06F 3/011
2017/0236328	12/2016	Eatedali	345/633	G01C 21/3688
2017/0237747	12/2016	Quinn	726/29	G06F 21/602
2017/0262855	12/2016	Venugopalan	N/A	G06Q 20/40145
2017/0345195	12/2016	Eatedali	N/A	G06Q 20/123
2017/0364920	12/2016	Anand	N/A	H04L 63/0861
2018/0059775	12/2017	Shafer	N/A	G06T 11/60
2018/0063088	12/2017	Hardy	N/A	H04L 63/0272

2018/0089898	12/2017	Huddy	N/A	H04W 4/029
2018/0104595	12/2017	Kawachiya et al.	N/A	N/A
2018/0109510	12/2017	Tommy	N/A	H04L 63/083
2018/0123816	12/2017	Tandon et al.	N/A	N/A
2018/0150690	12/2017	Yin	N/A	A61B 3/0008
2018/0285538	12/2017	Zhu	N/A	G06F 21/31
2018/0285549	12/2017	Sonkar	N/A	G06Q 20/40145
2018/0321744	12/2017	Cheaz	N/A	G06F 3/011
2019/0089701	12/2018	Mercury	N/A	G06Q 10/105
2019/0138914	12/2018	Meadows	N/A	H04L 51/52
2019/0171807	12/2018	Gadge	N/A	G06F 3/017
2019/0354669	12/2018	Brantley	N/A	G06F 21/36
2019/0369837	12/2018	Davis	N/A	H04N 7/157
2019/0377417	12/2018	Friedman	N/A	G06F 3/017
2020/0089855	12/2019	Chou	N/A	H04L 9/32
2020/0160345	12/2019	Damania	N/A	G06V 40/70
2020/0228524	12/2019	Szafranski	N/A	A63F 13/79
2020/0233485	12/2019	Chan	N/A	A61M 21/02
2020/0293120	12/2019	Timonen	N/A	G06F 3/011
2020/0338458	12/2019	Huang et al.	N/A	N/A
2020/0412538	12/2019	Rosado	N/A	H04L 63/126
2021/0004452	12/2020	Swaminathan	N/A	G06F 21/43
2021/0042748	12/2020	Sepulveda et al.	N/A	N/A
2021/0089637	12/2020	Cummins	N/A	H04L 63/0861
2021/0134110	12/2020	Irwin, Jr.	N/A	G06F 21/31
2021/0141880	12/2020	Pauli	N/A	G06F 3/013
2021/0209214	12/2020	Martin	N/A	G06F 3/013
2021/0256597	12/2020	Soppin	N/A	H04L 63/0853
2021/0318894	12/2020	Swaminathan	N/A	G06F 9/44526
2021/0328985	12/2020	Albero	N/A	G06F 21/34
2021/0358222	12/2020	Pejsa	N/A	G06F 3/0346
2022/0038891	12/2021	duPont	N/A	G06V 10/25
2022/0066621	12/2021	Appelbaum	N/A	H04L 65/1093
2022/0070167	12/2021	Adams, Jr.	N/A	G06F 21/30
2022/0116231	12/2021	Choi	N/A	G06Q 20/40145
2022/0167021	12/2021	French	N/A	H04N 21/2187
2022/0188833	12/2021	May	N/A	G06Q 20/40145
2022/0197989	12/2021	Rakshit	N/A	G06F 3/017
2022/0327225	12/2021	Lyren	N/A	H04S 1/007
2022/0374505	12/2021	Katz	N/A	G06F 3/013
2023/0026368	12/2022	Silverstein	N/A	G06F 16/986
2023/0085183	12/2022	Kuneva	705/72	G06Q 30/0601
2023/0128474	12/2022	Rodriguez Bravo	726/22	H04L 63/1416
2023/0136394	12/2022	Saito	726/6	G06F 21/31
2023/0139813	12/2022	Thiel	726/7	G06F 21/31
2023/0153884	12/2022	Parekh	705/26.7	G06Q 50/01
2023/0161887	12/2022	Albero	N/A	G06F 21/604
2023/0206781	12/2022	Robinson	434/43	A63F 13/803
2023/0254300	12/2022	Silverstein	726/10	G06F 21/30
2023/0291563	12/2022	Pan	N/A	H04L 9/001

2023/0291740	12/2022	Ashby	N/A	H04L 63/083
2023/0305621	12/2022	Hadi	N/A	G06F 3/012
2023/0316263	12/2022	Eby	N/A	H04L 9/3271
2023/0319030	12/2022	Young	726/6	G06F 3/0482
2023/0342760	12/2022	Patel	N/A	H04L 9/3234
2023/0370290	12/2022	Agarwal	N/A	H04L 9/3228
2023/0372830	12/2022	Gillis	N/A	H04L 9/3228
2023/0388299	12/2022	Gupta	N/A	H04L 63/0861
2023/0403270	12/2022	Kim	N/A	H04L 63/0861
2024/0004975	12/2023	Bivens	N/A	G06F 21/316
2024/0007464	12/2023	Mavani	N/A	H04L 63/0853
2024/0009574	12/2023	Horstman	N/A	A63F 13/71
2024/0012888	12/2023	Varma	N/A	H04L 67/306
2024/0013166	12/2023	Padiyar	N/A	G06Q 20/4014
2024/0022553	12/2023	Ingram	N/A	H04L 63/08
2024/0022561	12/2023	Ingram	N/A	H04L 63/08
2024/0022599	12/2023	Merdassa	N/A	H04L 63/1433
2024/0028675	12/2023	Merdassa	N/A	G06F 21/31
2024/0031162	12/2023	Singh	N/A	H04L 9/50
2024/0031182	12/2023	Krol	N/A	H04L 12/1831
2024/0031346	12/2023	Merdassa	N/A	H04L 63/08
2024/0036900	12/2023	Rudraraju	N/A	H04L 63/1483
2024/0037196	12/2023	Polasa	N/A	G06F 11/3438
2024/0037514	12/2023	Padiyar	N/A	G06Q 20/384
2024/0037861	12/2023	Gupta	N/A	H04L 9/3213
2024/0039926	12/2023	Gupta	N/A	H04L 67/131
2024/0039935	12/2023	Rudraraju	N/A	H04L 63/1425
2024/0070344	12/2023	Samuel	N/A	G06F 30/20
2024/0070653	12/2023	Cho	N/A	H04L 9/50
2024/0073205	12/2023	Gupta	N/A	H04L 63/10
2024/0078297	12/2023	Bhatia	N/A	G06F 3/011
2024/0080194	12/2023	Akkapeddi	N/A	H04L 63/107
2024/0086030	12/2023	Patel	N/A	G06F 3/04815
2024/0089327	12/2023	Patel	N/A	H04L 65/403
2024/0106803	12/2023	Race	N/A	H04L 9/008
2024/0111847	12/2023	Akkapeddi	N/A	G06F 21/31
2024/0112409	12/2023	Cross	N/A	G06T 19/003
2024/0118691	12/2023	Valeri	N/A	G05D 1/0016
2024/0118784	12/2023	Wang	N/A	G06F 3/0484
2024/0129302	12/2023	Singh	N/A	H04L 63/0853
2024/0135482	12/2023	Wang	N/A	G06F 3/011
2024/0135649	12/2023	Singh	N/A	A63F 13/87
2024/0137223	12/2023	Wang	N/A	H04L 9/3236
2024/0143709	12/2023	Pandey	N/A	G06F 21/31
2024/0143721	12/2023	Naik	N/A	G06T 19/006
2024/0152917	12/2023	Singh	N/A	H04L 9/50
2024/0157240	12/2023	Chayanam	N/A	A63F 13/55
2024/0160272	12/2023	Soryal	N/A	G06F 3/011
2024/0160704	12/2023	Albero	N/A	G06F 21/16
2024/0163284	12/2023	Albero	N/A	H04L 63/0861

2024/0169037	12/2023	Dravneek	N/A	G06F 21/31
2024/0171413	12/2023	Dravneek	N/A	H04L 9/3273
2024/0177413	12/2023	Padiyar	N/A	G06V 10/141
2024/0185514	12/2023	Singh	N/A	G06F 3/013
2024/0187235	12/2023	Brid	N/A	H04L 9/30
2024/0195825	12/2023	Durvasula	N/A	H04L 63/1425
2024/0221323	12/2023	Mukherjee	N/A	G06V 10/761
2024/0257101	12/2023	Rai	N/A	G06Q 30/0601
2024/0257466	12/2023	Harsha	N/A	G06T 17/205
2024/0259201	12/2023	Enciso	N/A	H04L 63/0428
2024/0273005	12/2023	Bussa	N/A	G06F 11/3692
2024/0273006	12/2023	Bussa	N/A	G06F 11/3692
2024/0273012	12/2023	Bussa	N/A	G06F 11/3698
2024/0275770	12/2023	Agrawal	N/A	H04L 63/08
2024/0283649	12/2023	Richter	N/A	H04L 9/3297
2024/0305647	12/2023	Rodriguez Bravo	N/A	H04L 63/1416
2024/0378280	12/2023	Brown	N/A	G06F 21/44
2025/0061635	12/2024	Dave	N/A	G09B 21/00

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Background/Summary

TECHNICAL FELD

(1) The present disclosure relates generally to network communication, and more specifically to a system and method for generating a virtual experience in a virtual environment.

BACKGROUND

(2) In a network environment, user devices are in data communication with other user devices that may be distributed anywhere in the world. These network environments allow data and information to be shared among these devices. Some of the technical challenges that occur when data is exchanged between devices are controlling data leakage, unauthorized access to data, and preventing malicious activities. Data storing user devices, such as computers, laptops, augmented reality devices, virtual reality devices, and smartphones, are vulnerable to attacks. This vulnerability poses several network security challenges. Existing systems are typically unable to detect a malicious attack until after the attack has occurred. For example, a bad actor may pretend to be another user in a virtual environment which then allows the bad actor to gain access to other users' information.

SUMMARY

(3) The system and methods implemented by the system as disclosed in the present disclosure provide technical solutions to the technical problems discussed above by allowing a user to securely access a virtual environment and perform secure data interactions in the virtual environment. The disclosed system and methods provide several practical applications and technical advantages.

(4) For example, the disclosed system and methods provide the practical application of improving interoperability of real-world systems and virtual world systems (e.g., metaverse systems) so that information may be seamlessly shared between these systems to implement data security, authorization of data interactions, and other data interactions performed in real-world and virtual environments. For example, user information collected from the user and/or assigned to the user in

a real-world environment may be used in a virtual environment (e.g., metaverse environment) to authenticate the user before allowing the user to access the virtual environment and perform any kind of action or interaction within the virtual environment. Additionally or alternatively, as described in embodiments of the present disclosure, user information collected from the user and/or assigned to the user in the real-world environment may be used in the virtual environment (e.g., metaverse environment) to provide access to products within the virtual environment. This process provides improved information security because it authenticates that an avatar is associated with the user and not an unauthorized party and that the user is authorized to access the virtual environment, and products provided in the virtual environment.

(5) Thus, the disclosed system and methods improve data security in the virtual environment. By improving data security in virtual environment, the disclosed system and methods generally improve technology related to performing secure data interactions in a virtual environment.

(6) The disclosed system and methods provide the additional practical application of saving memory resources. The seamless data flow between the real-world systems and virtual-world systems as a result of interoperability of these systems allows each system to store less data by avoiding the need to store the same data (e.g., authentication data, login credentials, etc.) in both systems, as data stored in one system can be accessed, or otherwise leveraged, by the other system. This saves memory resources by avoiding duplication of data. The saving of memory resources may leave more system memory for storing other data used in other technical operations. This provides the additional technical advantage of improving processing efficiency of computing systems that manage the real-world and virtual word environments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 is a schematic diagram of a system, in accordance with certain aspects of the present disclosure;

(3) FIG. 2 is a block diagram of an embodiment of the first user device used by the system of FIG. 1; and

(4) FIG. 3 illustrates a flowchart of an example method for performing data interactions in a virtual environment, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(5) Example System

(6) FIG. 1 is a schematic diagram of a system **100**, in accordance with certain aspects of the present disclosure. System **100** may include a first user device **104**, a second user device **106**, real-world server **130**, and a virtual-world server **150**, each connected to a network **180**. A first user **110** is associated with the first user device **104** and a second user **112** is associated with the second user device **106**. The system **100** may be communicatively coupled to the communication network **190** and may be operable to transmit data between each one of the first user device **104**, second user device **106**, real-world server **130**, and virtual-world server **150** through the communication network **190**.

(7) In general, the system **100** may improve interoperability of real-world systems and virtual world systems (e.g., metaverse systems) so that information may be seamlessly shared between these systems to implement data security, authorization of data interactions, access to virtual sub-environments and other data interactions performed in real-world and virtual environments. For example, user information collected from the user and/or assigned to the user in a real-world

environment may be used in a virtual environment **102** (e.g., metaverse environment) to authenticate the first user **110** before allowing the first user **110** to access the virtual environment **102** and perform any kind of action or interaction within the virtual environment **102**. Additionally or alternatively, user information collected from the first user **110** and/or assigned to the first user **110** in the real-world environment or virtual environment **102** may be used in the virtual environment **102** (e.g., metaverse environment) to provide the first user **110** access to products and/or services within the virtual environment **102**. This process provides improved information security because it authenticates that a first avatar **114** is associated with the first user **110**, not an unauthorized party, and that the first user **110** is authorized to access the virtual environment **102**, obtain products and services within the virtual environment **102** and perform data interactions. Further, a user may obtain a product by transferring real-world data objects between real-world entities based on data interactions performed in the virtual environment **102**.

(8) It may be noted that the terms “real-world” and “real-world environment” in this disclosure refer to any non-virtual environment where users (e.g., users **110** and **112**) can physically interact with real persons and objects. A real-world data interaction may refer to any data interaction performed outside the virtual environment **102** (e.g., a metaverse environment). Further, it may be noted that while certain embodiments of the present disclosure may be described in the context of a metaverse environment which is an example of a virtual environment **102**, the methods discussed in this disclosure apply to any other virtual environment **102**. The terms “virtual environment” and “metaverse environment” are used interchangeably throughout this disclosure. Furthermore, it may be noted that while certain embodiments of this disclosure describe one or more operations in relation to the first user **110**, these embodiments apply to any other user (e.g., second user **112**) connected to network **190**.

(9) The first user **110** may access the virtual environment **102** (e.g., a metaverse environment) through the first user device **104**. The first user device **104** is configured to display a two-dimensional (2D) or three-dimensional (3D) representation of the virtual environment **102** to the first user **110**. Examples of a virtual environment **102** may include, but are not limited to, a graphical or virtual representation of a metaverse, a map, a building interior, a landscape, a fictional location, an alternate reality, or any other suitable type of location or environment. The virtual environment **102** may be configured to use realistic or non-realistic physics for the motion of objects within the virtual environment **102**. For example, some virtual environments **102** may be configured to use gravity whereas other virtual environments **102** may not be configured to use gravity. Within the virtual environment **102**, each user may be associated with an avatar (such as the first avatar **114** for the first user **110**). An avatar is a graphical representation of a user at a virtual location within the virtual environment **102**. In embodiments, the virtual location of the avatar may be correlated to the physical location of a user in the real-world environment. Examples of an avatar may include, but are not limited to, a person, an animal, or an object. In some embodiments, the features and characteristics of the avatar may be customizable, and user defined. For example, the size, shape, color, attire, accessories, or any other suitable type of appearance features may be specified by a user. By using an avatar, a user is able to move within the virtual environment **102** to interact with one or more avatars and objects within the virtual environment **102** while independently remaining at a physical location in the real-world environment or being in transit in the real-world environment.

(10) While engaging in the virtual environment **102** via the first avatar **114**, the first user **110** may interact with a plurality of other users, objects and/or entities through a respective avatar. For example, the second user **112** may attempt to engage in an interaction session with the first avatar **114** through a second avatar **116** associated with the second user **112**. In another example, the first avatar **114** of the first user **110** may access a virtual sub-environment **120** within the virtual environment **102** and perform virtual data interactions within the virtual sub-environment **120**. In the real-world environment, the second user **112** may be physically located at a distance away from

the first user **110**. The second user **112** may access the virtual environment **102** through the second user device **106** to control the second avatar **116** and attempt to engage in an interaction session with the first user **110** through the first avatar **114**.

(11) Before the interaction between the first avatar **114** and the second avatar **116** occurs or the first avatar **114** can access the virtual sub-environment **120**, the virtual-world server **150** may authenticate that the first avatar **114** is associated with the first user **110** and not an unauthorized third-party. For example, the first user **110** may be required to sign into a secure portal that provides access to a data file (e.g., real-world data file **138** and/or virtual data file **160**) associated with the first user **110**. As shown in FIG. 1, a real-world data file **138** of the first user **110** is stored and managed by the real-world server **130** and a virtual data file **160** of the first user **110** is stored and managed by the virtual-world server **150**. In one or more embodiments, the virtual-world server **150** may employ single sign-on (SSO), multifactor authentication, or any other suitable authentication scheme in order to allow the first user **110** access to the virtual data file **160**. Similarly, the real-world server **130** may employ single sign-on (SSO), multifactor authentication, or any other suitable authentication scheme in order to allow the first user **110** access to the real-world data file **138**. The virtual data file **160** and the real-world data file **138** may include virtual data objects **162** and real-world data objects **140** respectively owned by the first user **110**. The real-world server **130** and the virtual-world server **150** may store other information related to the first user **110** including, but not limited to, user profile information, account information (e.g., including identity and other details relating to the respective data files **138** and **160**), avatar information, digital assets (e.g., respective real-world data objects **140** and virtual data objects **162**) information, or any other suitable type of information that is associated with a user within the virtual environment **102** and/or the real-world environment.

(12) As shown in FIG. 1, virtual-world server **150** comprises a processor **152**, a memory **156**, and a network interface **154**. The processor **152** comprises one or more processors operably coupled to the memory **156**. The processor **152** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g. a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **152** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **152** is communicatively coupled to and in signal communication with the memory **156**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **152** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **152** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(13) The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions (e.g., virtual-world server instructions **166**) to implement the virtual-world server **150**. In this way, processor **152** may be a special-purpose computer designed to implement the functions disclosed herein. In one or more embodiments, the virtual-world server **150** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The virtual-world server **150** is configured to operate as described with reference to FIGS. 1 and 3. For example, the processor **152** may be configured to perform at least a portion of the method **300** as described in FIG. 3.

(14) The memory **156** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **156** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory

(DRAM), and static random-access memory (SRAM).

(15) The memory **156** is operable to store user credential **170**, user profile **172**, experience themes **174**, product types **176**, products **178**, virtual data file **160**, virtual data objects **162**, benefit counts **164** and the virtual-world server instructions **166**. The virtual-world server instructions **166** may include any suitable set of instructions, logic, rules, or code operable to execute the virtual-world server **150**.

(16) The network interface **154** is configured to enable wired and/or wireless communications. The network interface **154** is configured to communicate data between the virtual-world server **150** and other devices, systems, or domains (e.g. user devices **104** and **106** and real-world server **130**). For example, the network interface **154** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **152** is configured to send and receive data using the network interface **154**. The network interface **154** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(17) The real-world server **130** comprises a processor **132**, a memory **134**, and a network interface **136**. The processor **132** comprises one or more processors operably coupled to the memory **134**. The processor **132** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g. a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **132** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **132** is communicatively coupled to and in signal communication with the memory **134**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **132** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **132** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(18) The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions (e.g., real-world server instructions **146**) to implement the real-world server **130**. In this way, processor **132** may be a special-purpose computer designed to implement the functions disclosed herein. In one or more embodiments, the real-world server **130** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The real-world server **130** is configured to operate as described with reference to FIGS. **1** and **3**. For example, the processor **132** may be configured to perform at least a portion of the method **300** as described in FIG. **3**.

(19) The memory **134** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **134** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(20) The memory **134** is operable to store information relating to real-world data file **138**, real-world data objects **140**, benefit counts **142** and the real-world server instructions **146**. The real-world server instructions **146** may include any suitable set of instructions, logic, rules, or code operable to execute the real-world server **130**.

(21) The network interface **136** is configured to enable wired and/or wireless communications. The network interface **136** is configured to communicate data between the real-world server **130** and other devices, systems, or domains (e.g. user devices **104** and **106**, and virtual-world server **150**). For example, the network interface **136** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **132** is configured to send and receive data

using the network interface **136**. The network interface **136** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(22) Each of the real-world server **130** and the virtual-world server **150** is generally a suitable server (e.g., including a physical server and/or virtual server) operable to store data in a memory (e.g., respective memories **134** and **156**) and/or provide access to application(s) or other services. One or both of the real-world server **130** and the virtual-world server **150** may be a backend server associated with a particular entity (e.g., organization) that facilitates conducting data interactions between entities, between one or more users, and/or between a user and an entity. In other embodiments, one or both of the real-world server **130** and the virtual-world server **150** may be organized in a distributed manner, or by leveraging cloud computing technologies. Real-world server **130** may store information which is primarily used to support data interactions performed in the real-world environment. Virtual-world server **150** may store information which is primarily used to support data interactions performed in the virtual environment **102** (e.g., a metaverse environment). It may be noted that the operations performed by the real-world server **130** and the virtual-world server **150** described in embodiments of the present disclosure may be implemented by a single server.

(23) The communication network **190** may facilitate communication within the system **100**. This disclosure contemplates the communication network **190** being any suitable network operable to facilitate communication between the first user device **104**, second user device **106**, real-world server **130**, and virtual-world server **150**. Communication network **190** may include any interconnecting system capable of transmitting audio, video, signals, data, messages, or any combination of the preceding. Communication network **190** may include all or a portion of a local area network (LAN), a wide area network (WAN), an overlay network, a software-defined network (SDN), a virtual private network (VPN), a packet data network (e.g., the Internet), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a Plain Old Telephone (POT) network, a wireless data network (e.g., WiFi, WiGig, WiMax, etc.), a Long Term Evolution (LTE) network, a Universal Mobile Telecommunications System (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a Near Field Communication network, a Zigbee network, and/or any other suitable network, operable to facilitate communication between the components of system **100**. In other embodiments, system **100** may not have all of these components and/or may have other elements instead of, or in addition to, those above.

(24) Each of the user devices (i.e., first user device **104** and second user device **106**) may be any computing device configured to communicate with other devices, such as a server (e.g., real-world server **130**, virtual-world server **150**), databases, etc. through the communication network **190**. Each of the user devices may be configured to perform specific functions described herein and interact with one or both of real-world server **130** and the virtual-world server **150**, e.g., via respective user interfaces. Each of the user devices is a hardware device that is generally configured to provide hardware and software resources to a user. Examples of a user device include, but are not limited to, a virtual reality device, an augmented reality device, a laptop, a computer, a smartphone, a tablet, a smart device, an Internet-of-Things (IoT) device, or any other suitable type of device. The user devices may comprise a graphical user interface (e.g., a display), a touchscreen, a touchpad, keys, buttons, a mouse, or any other suitable type of hardware that allows a user to view data and/or to provide inputs into the user device. Each user device may be configured to allow a user to send requests to one or both of real-world server **130** and the virtual-world server **150**, or to another user device.

(25) Example User Device

(26) FIG. 2 is a block diagram of an embodiment of the first user device **104** used by the system of FIG. 1. First user device **104** may be configured to display the virtual environment **102** (referring to FIG. 1) within a field of view of the first user **110** (referring to FIG. 1), capture biometric, sensory, and/or physical information of the first user **110** wearing the first user device **104**, and to facilitate

an electronic interaction between the first user **110** and the second user **112** (referring to FIG. **1**) or between the first user **110** and an entity (e.g., represented by a virtual entity in the virtual environment **102**).

(27) First user device **104** comprises a processor **202**, a memory **204**, and a display **206**. Further embodiments may include a camera **208**, a wireless communication interface **210**, a network interface **212**, a microphone **214**, a global position system (GPS) sensor **216**, and/or one or more biometric devices **218**. First user device **104** may be configured as shown or in any other suitable configuration. For example, first user device **104** may comprise one or more additional components and/or one or more shown components may be omitted.

(28) The processor **202** comprises one or more processors operably coupled to and in signal communication with memory **204**, display **206**, camera **208**, wireless communication interface **210**, network interface **212**, microphone **214**, GPS sensor **216**, and biometric devices **218**. Processor **202** is configured to receive and transmit electrical signals among one or more of memory **204**, display **206**, camera **208**, wireless communication interface **210**, network interface **212**, microphone **214**, GPS sensor **216**, and biometric devices **218**. The electrical signals are used to send and receive data (e.g., images captured from camera **208**, virtual objects to display on display **206**, etc.) and/or to control or communicate with other devices. Processor **202** may be operably coupled to one or more other devices (for example, the real-world server **130** and/or virtual-world server **150** shown in FIG. **1**).

(29) The processor **202** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **202** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **202** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **202** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(30) The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions to implement the function disclosed herein, such as some or all of those described with respect to FIGS. **1** and **3**. For example, processor **202** may be configured to display virtual objects on display **206**, detect hand gestures, identify virtual objects selected by a detected hand gesture, capture biometric information of a user, such as first user **110**, via one or more of camera **208**, microphone **214**, and/or biometric devices **218**, and communicate via wireless communication interface **210** with the real-world server **130**, virtual-world server **150** and/or second user device **106**. In some embodiments, the function described herein is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware or electronic circuitry.

(31) The memory **204** is operable to store any of the information described with respect to FIGS. **1** and **3** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor **202**. For example, the memory **204** may store the instructions **220**. The memory **204** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. Memory **204** is operable to store, for example, information relating to the identity of the user (e.g., at least a portion of user profile **172**), instructions for performing the functions of first user device **104** described herein, and any other data or instructions. The memory **204** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory

(RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(32) Display **206** is configured to present visual information to a user (for example, first user **110** in FIG. **1**) in a virtual reality environment, an augmented reality environment or mixed reality environment. In other embodiments, the display **206** is configured to present visual information to the user as the virtual environment **102** (referring to FIG. **1**) in real-time. In an embodiment, display **206** is a wearable optical display (e.g., glasses or a headset) configured to reflect projected images and enables a user to see through the display. For example, display **206** may comprise display units, lens, semi-transparent mirrors embedded in an eye glass structure, a visor structure, or a helmet structure. Examples of display units include, but are not limited to, a cathode ray tube (CRT) display, a liquid crystal display (LCD), a liquid crystal on silicon (LCOS) display, a light emitting diode (LED) display, an active-matrix OLED (AMOLED), an organic LED (OLED) display, a projector display, or any other suitable type of display as would be appreciated by one of ordinary skill in the art upon viewing this disclosure. In another embodiment, display **206** is a graphical display on a user device. For example, the graphical display may be the display of a tablet or smart phone configured to display virtual environment **102**.

(33) Examples of camera **208** include, but are not limited to, charge-coupled device (CCD) cameras and complementary metal-oxide semiconductor (CMOS) cameras. Camera **208** is configured to capture images of a wearer of first user device **104**, such as first user **110**. Camera **208** may be configured to capture images continuously, at predetermined intervals, or on-demand. For example, camera **208** may be configured to receive a command from first user **110** to capture an image. In another example, camera **208** is configured to continuously capture images to form a video stream. Camera **208** is communicably coupled to processor **202**.

(34) Examples of wireless communication interface **210** include, but are not limited to, a Bluetooth interface, an RFID interface, a near field communication interface, a local area network (LAN) interface, a personal area network interface, a wide area network (WAN) interface, a Wi-Fi interface, a ZigBee interface, or any other suitable wireless communication interface as would be appreciated by one of ordinary skill in the art upon viewing this disclosure. Wireless communication interface **210** is configured to facilitate processor **202** in communicating with other devices. For example, wireless communication interface **210** is configured to enable processor **202** to send and receive signals with other devices, such as second user device **106**, the real-world server **130** and/or virtual-world server **150** (referring to FIG. **1**). Wireless communication interface **210** is configured to employ any suitable communication protocol.

(35) The network interface **212** is configured to enable wired and/or wireless communications. The network interface **212** is configured to communicate data between the first user device **104** and other network devices, systems, or domain(s). For example, the network interface **212** may comprise a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. The processor **202** is configured to send and receive data using the network interface **212**. The network interface **212** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(36) Microphone **214** is configured to capture audio signals (e.g., voice signals or commands) from a user, such as first user **110**. Microphone **214** is configured to capture audio signals continuously, at predetermined intervals, or on-demand. Microphone **214** is communicably coupled to processor **202**.

(37) GPS sensor **216** is configured to capture and to provide geographical location information. For example, GPS sensor **216** is configured to provide a geographic location of a user, such as first user **110**, employing first user device **104**. GPS sensor **216** may be configured to provide the geographic location information as a relative geographic location or an absolute geographic location. GPS sensor **216** may provide the geographic location information using geographic coordinates (i.e., longitude and latitude) or any other suitable coordinate system. GPS sensor **216** is communicably

coupled to processor **202**.

(38) Examples of biometric devices **218** may include, but are not limited to, retina scanners, fingerprint scanners and facial scanners. Biometric devices **218** are configured to capture information about a person's physical characteristics and to output a biometric signal based on captured information. A biometric signal is a signal that is uniquely linked to a person based on their physical characteristics. For example, biometric device **218** may be configured to perform a retinal scan of the user's eye and to generate a biometric signal for the user based on the retinal scan. As another example, a biometric device **218** is configured to perform a fingerprint scan of the user's finger and to generate a biometric signal for the user based on the fingerprint scan. Biometric device **218** is communicably coupled to processor **202**.

(39) Referring back to FIG. **1**, in one or more embodiments, one or both of the real-world server **130** and the virtual-world server **150**, and one or more user devices (e.g., second user device **106**) may be part of an Information Technology (IT) infrastructure of an entity or organization. For example, second user **112** may be a representative of the entity or organization who may use the second user device **106** to enter the virtual environment **102** and virtually interact with one or more users (e.g., first user **110**) via the second avatar **116** to provide services to the first user **110**.

(40) The real-world server **130** may be configured to allow users (e.g., first user **110**) registered with the real-world server **130** to perform one or more data interactions in the real-world environment. Similarly, virtual-world server **150** may be configured to allow users (e.g., first user **110**) registered with the virtual-world server **150** to perform one or more data interactions in the virtual environment **102** (e.g., a metaverse environment). When the real-world server **130** and/or the virtual-world server **150** are owned and/or operated by a particular entity or organization (e.g., are part of the IT infrastructure of the entity or organization), being registered with the real-world server **130** and/or the virtual-world server **150** may also be interpreted as being registered with the particular entity or organization. For example, when the first user **110** is registered with the virtual-world server **150**, this may be interpreted as the first user **110** being registered with the entity or organization that owns and/or manages the virtual world server **150**. In one embodiment, the real-world server **130** and the virtual-world server **150** are owned and/or operated by the same entity/organization. In this context, virtual-world server **150** may be configured to allow users (e.g., first user **110**) registered with the real-world server **130** to perform one or more data interactions in the virtual environment **102** (e.g., a metaverse environment). In alternative embodiments, the real-world server **130** and the virtual-world server **150** may be owned and/or operated by different entities/organizations. For example, the real-world server **130** may be owned and/or operated by a first entity and the virtual-world server **150** may be owned and/or operated by a second entity different from the first entity.

(41) In one or more embodiments, as the first user **110** initially registers with the real-world server **130** in the real-world environment, the real-world server **130** may collect several pieces of information from the user including information relating to the identity of the user such as legal name, social security number, biometrics (e.g., fingerprints, retina scans, face ID etc.), residence address, phone numbers, assets owned by the user, and copies of government issued documents (e.g., drivers permit, state identity card etc.). This information is stored by real-world server **130** as part of user profile **172** of the first user **110**. In one embodiment, at least a portion of the user profile **172** relating to the first user **110** collected in the real-world environment may be stored in the virtual-world server **150**. Once the identity of the first user **110** is confirmed and all other information provided by the first user **110** is verified to be correct, the real-world server **130** may generate a real-world data file **138** for the first user **110** in which the first user **110** may store real-world data objects **140** owned by the first user **110**. In one example, the first user **110** may engage in a real-world interaction with a service representative managing the real-world server **130** (e.g., physical interaction at an office location, over phone, voice chat etc.) to provide such information that can be used to register the first user **110** at the real-world server **130** and generate the real-

world data file **138** of the first user **110**. In another example, the first user **110** may engage in a real-world interaction by accessing a webpage provided and managed by the real-world server **130**. Once the first user **110** initiates a registration process via the webpage, the real-world server **130** may walk the first user **110** through several steps in which the first user **110** may be requested to provide information necessary to verify the identity of the first user **110** and register the first user **110** with the real-world server **130**.

(42) Information relating to the real-world data file **138** of the first user **110** may be stored as part of the user profile **172** of the first user **110**. This information may include, but is not limited to, an identity (e.g., unique name or number) of the real-world data file **138**, amount of real-world data objects **140** stored in the real-world data file **138**, a log of data interactions conducted in relation to the real-world data file **138** and any other information relating to the real-world data file **138**. As described below, the information relating to the real-world data file **138** of the first user **110** may further include a benefit count **142** associated with the real-world data file **138**.

(43) Once registered with the real-world server **130**, the real-world server **130** may allow the first user **110** to perform one or more data interactions in the real-world environment. For example, a real-world data interaction may include transferring one or more real-world data objects **140** from the real-world data file **138** of the first user **110** to a second real-world data file (not shown) of the second user **112** or another entity. Another example data interaction may include receiving one or more real-world data objects **140** in the real-world data file **138** of the first user **110** from the second real-world data file of the second user **112** or another entity. Another example data interaction may include requesting by the first user **110** transfer of real-world data objects from a data file of a second user to a user data file of a third user as part of satisfying an agreement between the first user **110** and the third user. Another example data interaction may include modifying at least a portion of the user profile **172** (e.g., user credentials to access the real-world server, phone numbers, residential address, email address, information relating to user assets etc.) stored at the real-world server **130**. It may be noted that a data interaction in accordance with embodiments of the present disclosure refers to any interaction in the real-world environment and/or virtual environment **102** that includes transfer of data between computing nodes (e.g., first user device **104**, second user device **106**, real-world server **130** and virtual-world server **150**).

(44) The first user **110** may additionally register with the virtual-world server **150**. In one embodiment, when initially registering with the virtual-world server **150**, the first user **110** may provide to the virtual-world server **150** a credential (e.g., username and password) that provides the first user **110** access to the real-world server **130**. In one embodiment, a single web page or web portal may allow the first user **110** to register with the real-world server **130** as well as the virtual-world server **150**. The first user **110** may first register with the real-world server **130** as described above and generate credentials that allow the first user **110** access to the real-world server **130** and services provided by the real-world server **130**. Once registered with the real-world server **130**, the web portal may provide the first user **110** an option to additionally register with the virtual-world server **150** which may allow the first user **110** to perform data interactions in the virtual environment **102**. Registration with the virtual-world server **150** may include generating a user credential **170** that allows the first user **110** to log in to the virtual-world server **150** and enter the virtual environment **102** via first avatar **114** of the first user **110**. Once registered with the virtual-world server **150**, the first user **110** may generate a virtual data file **160** in which the first user **110** may store virtual data objects **162** owned by the first user **110**. In one or more embodiments, the virtual data file **160** of the first user **110** is associated with the real-world data file **138** of the first user **110**. For example, the virtual data file **160** is a virtual image of the real-world data file **138**, wherein the virtual data objects **162** correspond to the real-world data objects **140**. In other words, the virtual data file **160** is a virtual representation of the real-world data file **138**. In another example, the virtual data file **160** stores a portion of the real-world data objects **140** in the form of virtual data objects **162**. In another example, real-world data objects **140** may be converted to

virtual data objects **162**, and vice versa. In this case, there may not be a one-to-one conversion between the real-world data objects **140** and virtual data objects **162**. For example, one real-world data object **140** may be converted to a plurality of virtual data objects **162**, wherein the conversion ratio may dynamically change from time to time. In one embodiment, when the real-world server **130** and the virtual-world server **150** are owned and/or managed by the same entity or organization, the first user **110** may use the same user credential **170** to login to the real-world server **130** as well as the virtual-world server **150**.

(45) Information relating to the virtual data file **160** of the first user **110** may be stored by the virtual-world server **150** as part of the user profile **172** of the first user stored at the virtual-world server **150**. This information may include, but is not limited to, an identity of the virtual data file **160**, amount of virtual data objects **162** stored in the virtual data file **160**, a log of virtual data interactions conducted in the virtual environment **102** in relation to the virtual data file **160** and any other information relating to the virtual data file **160**. As described below, the information relating to the virtual data file **160** of the first user **110** may further include a benefit count **164** associated with the virtual data file **160**.

(46) In certain embodiments, when the real-world server **130** and the virtual-world server **150** are owned and/or managed by different entities, first user **110** may separately register with one of the real-world server **130** and the virtual-world server **150** without registering with the other. For example, when the real-world server **130** is registered with a first entity and the virtual-world server **150** is registered with a second entity different from the first entity, first user **110** may separately register with the virtual-world server **150** without registering or needing to register with the real-world server **130**. For example, as the first user **110** separately registers with the virtual-world server **150**, the virtual-world server **150** may collect several pieces of information from the first user **110** including information relating to the identity of the user such as legal name, social security number, biometrics (e.g., fingerprints, retina scans, face ID etc.), residence address, phone numbers, assets owned by the user, and copies of government issued documents (e.g., drivers permit, state identity card etc.). This information is stored by virtual-world server **150** as part of user profile **172** of the first user **110**. In one embodiment, at least a portion of the user profile **172** relating to the first user **110** collected in by the virtual-world server **150** may be stored in the real-world server **130**. Once the identity of the first user **110** is confirmed and all other information provided by the first user **110** is verified to be correct, the virtual-world server **150** may generate the virtual data file **160** for the first user **110** in which the first user **110** may store virtual data objects **162** owned by the first user **110**.

(47) Once registered with the virtual-world server **150**, the virtual-world server **150** may allow the first user **110** to perform one or more virtual data interactions. For example, a virtual data interaction may include transferring one or more virtual data objects **162** from the virtual data file **160** of the first user **110** to a second virtual data file (not shown) of the second user **112** or another entity. Another example data interaction may include receiving one or more virtual data objects **162** in the virtual data file **160** of the first user **110** from the second virtual data file of the second user **112** or another entity. Another example data interaction may include requesting by the first user **110** transfer of virtual data objects **162** from a data file of a second user to a data file of a third user as part of satisfying an agreement between the first user **110** and the third user. Another example data interaction may include modifying at least a portion of the user profile **172** (e.g., user credentials to access the real-world server, phone numbers, residential address, email address, information relating to user assets etc.) stored at the virtual-world server **150**.

(48) In one or more embodiments, the virtual data file (e.g. virtual data file **160**) is a software application running on a computing node owned and/or operated by the respective user (e.g., first user **110**). For example, when the first user **110** desires to receive virtual data objects **162** from a virtual data file of the second user **112**, first user **110** may direct the second user **112** to a unique cryptographic address (e.g., public key) issued by the virtual data file **160**. In one embodiment, the

virtual data file **160** may not itself store the virtual data objects **162** but may store information that points to a location of the virtual data objects **162**, for example, on a server (e.g., virtual-world server **150**). Virtual data file **160** may be web-based or hardware-based. For example, virtual data file **160** may be stored in a mobile device or a desktop computer connected to the internet. Additionally or alternatively, virtual data file **160** may be stored in a device (e.g., USB drive) that is not connected to the network **190**.

(49) In certain embodiments, virtual environment **102** may include one or more virtual sub-environments **120**. A virtual sub-environment **120** may be a designated region within the virtual environment **102** for use by a user (e.g., first user **110**) or an entity/organization. For example, a virtual sub environment **120** may be assigned to a particular entity which may provide one or more services to users of the virtual environment **102**. The virtual sub-environment **120** may take any form in the virtual environment **102** including, but not limited to, a virtual room, a virtual building or portions thereof, a virtual store, or any other designated virtual space/region within the virtual environment **102**. Virtual-world server **150** may be configured to generate virtual sub-environments **120** with the virtual environment **102**. A virtual sub-environment **120** is usually generated for a customized purpose. For example, a virtual sub-environment **120** may be used by an entity that is assigned the virtual sub-environment **120** for receiving virtual data objects **162** from users for storing in virtual data files **160** and/or for distribution to specified one or more target entities. Further a virtual sub-environment may allow users (e.g., first user **110**) of the virtual environment **102** to perform virtual data interactions including transferring virtual data objects **162** to other users (e.g., second user **112**) and/or one or more target entities. As described below, virtual-world server **150** may be configured to generate customized virtual sub-environments **120** that may be used to provide customized virtual experiences to users.

(50) Data security is important in any system that supports online data interactions between computing nodes of the system. Online data interactions in real-world environments have existed for several decades and robust measures are already in place to ensure data security in real-world systems. However, virtual-world technology (e.g., metaverse technology) is relatively new and data security is a challenge in metaverse systems as the development of metaverse related technologies is still at a nascent stage and standardized systems that provide robust data security are not yet in place.

(51) Embodiments of the present disclosure leverage user data collected and/or assigned in a real-world environment for providing data security in a metaverse environment. In general, embodiments of the present disclosure describe techniques that improve interoperability of real-world systems and virtual world systems (e.g., metaverse systems) so that information may be seamlessly shared between these systems to implement data security and other data interactions performed in real-world and virtual environments. In one or more embodiments, the system and methods discussed herein use information stored in a user profile **172** of the first user **110** generated during real-world data interactions with the first user **110** to allow the first user **110** to securely obtain products and/or services in the virtual environment **102**.

(52) In one or more embodiments, virtual-world server **150** may be configured to use a user credential **170** collected from the first user **110**, generated by the first user **110** or assigned to the first user **110** during real-world data interactions with the first user **110**, to verify identity of the first user **110** in the virtual environment **102**. Thus, the user credential **170** provides the first user **110** access to the virtual environment **102**. For example, the user credential **170** may be used by the virtual-world server **150** to verify that the first avatar **114** belongs to and is controlled by the first user **110**.

(53) In one example, user credential **170** may include a username and password generated by the first user **110** as part of registering with the real-world server **130**. The virtual-world server **150** may allow the first user **110** to use the same username and password to enter the virtual environment **102** via first avatar **114**. In an alternative embodiment, the first user **110** may generate

a separate user credential (different from the user credential **170**) as part of registering with the real-world server **130** and use the separate user credential to log into the real-world server **130** and perform real-world data interaction. The first user **110** may separately generate user credential **170** as part of registering with the virtual-world server **150** (e.g., during a real-world data interaction with the real-world server **130**). Once the first user **110** has registered with the virtual-world server **150** and generated the user credential **170**, the first user **110** may use the user credential **170** to log into the virtual-world server **150** and perform virtual-world data interactions. The user credential **170** of the first user **110** may be associated with the user profile **172** of the first user **110**.

(54) In one embodiment, a second user **112** may gain access to the virtual-world server **150** and enter the virtual environment **102** without registering with the virtual-world server **150** and/or real-world server **130**. In other words, virtual-world server **150** may allow the second user **112** to enter the virtual environment **102** as a guest user. For example, in order to enter the virtual environment **102** as a guest user, second user **112** may need to complete a simple verification process that may include verifying the identity of the second user **112** (e.g., using email verification, text verification etc.) and setting up a temporary avatar (e.g., second avatar **116**) to navigate the virtual environment **102**. In one embodiment, if the identity of the second user **112** is successfully verified, virtual world server **150** may generate a temporary user credential **170** which the second user **112** may use to access the virtual environment **102** as a guest user. It may be noted that, a guest user who is not registered with the real-world server **130** or virtual-world server **150** may not have access to the full array of products and/or services provided by the respective real-world server **130** or virtual-world server **150**.

(55) Virtual-world server **150** may be configured to facilitate a user (e.g., first user **110**) to obtain one or more products in the virtual environment **102** or real-world environment. As shown in FIG. **1**, virtual-world server **150** may be configured to store one or more experience themes **174**, wherein each experience theme **174** relates to a particular type of experience that can be had by the user **110** in the virtual environment **102** or the real-world environment. An example of an experience theme associated with the virtual environment **102** may include visual elements that make up the virtual environment **102**. An example of an experience theme associated with the real-world environment may include a travel experience. An experience theme **174** includes one or more product types **176**, wherein each product type **176** is a type of product **178** that relates to the experience theme **174**. For example, when the experience theme **174** relates to visual experience of the first user **110** within the virtual environment **102**, example product types **176** associated with this experience theme **174** may include, but are not limited to, avatars, buildings, roads, objects, colors and patterns. In another example, when the experience theme **174** relates to a travel experience in the real-world environment, product types **176** associated with this experience theme **174** may include, but are not limited to, cars, buses, trains, travel destinations, hotels, restaurants, and excursions. An experience theme **174** may include several products **178** associated with each product type **176**. For example, when the experience theme **174** relates to a visual experience within the virtual environment **102**, the “avatars” product type **176** may include several specific avatars **114** that may be used as the first avatar **114** of the first user **110** within the virtual environment **102**. The “buildings” product type **176** may include a plurality of specific virtual buildings that can be used to generate the virtual environment **102**. The “colors” and “patterns” product type **176** may include a plurality of colors and patterns respectively that can be used to render specific objects or portions of the virtual environment **102**. When the experience theme **174** relates to a travel experience in the real-world environment, the “destination” product type **176** may include a selection of different destinations, the “cars” product type **176** may include a selection of different cars that be obtained for a travel, the “hotels” product type may include a selection of hotels at the selected destination and the like.

(56) In operation, when the first user **110** enters the virtual environment (e.g., using first avatar **114**) after logging in to the virtual-world server **150** by providing user credential **170**, the first user

110 may navigate to a virtual sub-environment **120** that is configured to facilitate users to obtain products **178** associated with one or more experience themes **174**. The virtual sub-environment **120** may be assigned to a particular real-world entity. Once the first user **110** has entered the virtual sub-environment **120**, the virtual-world server **150** may present to the first user **110** one or more experience themes **174**. For example, the virtual-world server **150** may display the experience themes **174** on a virtual display within the virtual sub-environment **120**. Virtual-world server **150** may be configured to allow the first user **110** to select one of the presented experience themes **174**. The first user **110** may select one of the experience themes **174** depending on what kind of experience the first user **110** is looking for. For example, the virtual-world server **150** may present a visual experience theme associated with the virtual environment **102** and a travel experience theme associated with the real-world environment. In this example, the first user **110** selects the visual experience theme associated with the virtual environment **102**. Certain embodiments of the present disclosure will now be described in accordance with an example in which the first user **110** selects the visual experience theme as an experience theme **174**. However, it may be noted that these embodiments apply in relation to other experience themes **174**.

(57) Once the virtual-world server **150** receives a selection of an experience theme **174**, virtual-world server **150** may be configured to present to the first user **110** within the virtual sub-environment **120**, one or more product types **176** associated with the selected experience theme **174**. For example, when the first user **110** selects the visual experience theme as the selected experience theme **174**, virtual-world server **150** presents a list of product types **176** associated with the visual experience theme including, but not limited to, avatars, buildings, roads, objects, colors and patterns. Virtual-world server **150** may be configured to allow the first user **110** to select one or more of the presented product types **176** associated with the selected experience theme **174**. As the first user **110** selects each product type **176**, virtual-world server **150** may be configured to access from the memory **156** one or more products **178** associated with the selected product type **176** and present the product **178** to the first user **110** in the virtual environment **102**. For example, when the first user **110** selects the “avatars” product type **177**, virtual-world server **150** may access from the memory **156** one or more specific avatars associated with the “avatars” product type **176** and present the one or more specific avatars to the first user **110** in the virtual-sub environment **120**. For example, virtual-world server **150** may display a 2-dimensional or 3-dimensional image of one or more avatars associated with the avatar product type **176**. Virtual-world server **150** may allow the first user **110** to select at least one product **178** (e.g., specific avatar) associated with a selected product type **176**. In one embodiment, virtual-world server **150** may allow the first user **110** to select multiple product types **176** related to the selected experience theme **174** and one or more products **178** associated with each of the multiple selected product types **176**. For example, the first user **110** may want to build a customized virtual sub-environment **120** and may want to obtain specific visual elements (e.g., avatars, buildings, roads, objects, colors, patterns etc.) to build the virtual sub-environment **120**. For this purpose, the first user **110** may select a specific avatar from the “avatar” product type **176**, one or more specific buildings from the “buildings” product type **176**, one or more roads from the “roads” product type **176**, one or more colors from the “colors” product type **176**, and one or more patterns from the “patterns” product type **176**.

(58) Once the product selections have been made by the first user **110**, virtual-world server **150** may be configured to determine one or more data files (e.g., real-world data files **138** or virtual data files **160**) that can be used to obtain at least one of the products **178** selected by the first user **110**. In this context, virtual-world server **150** may be configured to store information relating to a plurality of real-world data files **138** and virtual data files **160** that may be used by the first user **110** to obtain products **178** associated with experience themes **174**. A particular product **178** may be obtained by the first user **110** by transferring a pre-configured amount of real-world data objects **140** from a real-world data file **138** registered to the first user **110** to another real-world data file **138** registered to an entity that provides the particular product **178**, and/or by transferring a pre-

configured amount of virtual data objects **162** from a virtual data file **160** registered to the first user **110** to another virtual data file **160** registered to the entity that provides the particular product **178**. Different products **178** may be obtained by transferring different pre-configured amounts of real-world data objects **140** or virtual data objects **162** to respective data files of entities that provide the products **178**. For example, a first product **178** may be obtained by the first user **110** by transferring a first pre-configured amount of real-world data objects **140** or virtual data objects **162**, and a second product **178** may be obtained by the first user **110** by transferring a second pre-configured amount of real-world data objects **140** or virtual data objects **162**.

(59) In some embodiments, a single data file (e.g., real-world data file **138** or virtual data file **160**) may be used to obtain multiple products **178**. In some embodiments, any one of multiple data files (e.g., multiple real-world data file **138**, multiple virtual data files **160** or combinations thereof) may be used to obtain a same product **178**.

(60) In one or more embodiments, a data file (e.g., real-world data file **138** or virtual data file **160**) may provide a benefit count **142/164** to the first user **110**, when the first user **110** uses the data file (e.g., real-world data file **138** or virtual data file **160**) to obtain a particular product **178** (e.g., by transferring data objects to a respective data file of an entity that provides the particular product **178**). For example, the first user **110** may receive a pre-configured benefit count **142** when the first user **110** uses a particular real-world data file **138** to obtain a particular product **178**. Similarly, the first user **110** may receive a pre-configured benefit count **164** when the first user **110** uses a particular virtual data file **160** to obtain a particular product **178**. In one embodiment, different data files (e.g., different real-world data files **138** and/or different virtual data files **160**) may provide different benefit counts **142/164** to obtain a same product **178**. In an additional or alternative embodiment, a single data file (e.g., real-world data file **138** or virtual data file **160**) may provide different benefit counts **132/164** to obtain different products **178**.

(61) In one or more embodiments, benefit count **142** associated with a real-world data file **138** may be used to obtain one or more products **178** in the real-world environment, and benefit count **164** associated with a virtual data file **160** may be used to obtain one or more products **178** in the virtual environment **102**. A higher benefit count **142/164** generally allows the first user **110** to obtain more products **178**. Thus, a higher benefit count **142/164** is generally of higher value. The first user **110** may accumulate benefit counts **142/164** received as a result of using a particular data file (e.g., real-world data file **138** or virtual data file **160**) to obtain several products **178**.

(62) The information stored by the virtual-world server **150** in memory **156** relating to a plurality of real-world data files **138** and virtual data files **160** that may be used to obtain products **178** associated with experience themes **174** may include one or more of an identity of each real-world data file **138** or virtual data file **160**, products **178** that may be obtained using the real-world data file **138** or the virtual data file **160**, and benefit counts **142** or **164** provided by the real-world data file **138** or virtual data file **160** respectively when used to obtain each of the products **178**.

(63) As described above, once the selection of the products **178** associated with a selected experience theme **174** has been made by the first user **110**, virtual-world server **150** may be configured to determine one or more data files (e.g., real-world data files **138** or virtual data files **160**) that can be used to obtain one or more of the products **178** selected by the first user **110**. For a selected product **178**, virtual-world server **150** determines one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that may be used to obtain the product **178**, and a benefit count **142/164** corresponding to each of the one or more data files that the first user **110** may receive when the data file (e.g., real-world data file **138** or virtual data file **160**) is used to obtain the product **178**. For example, based on the selected product **178**, virtual-world server **150** may be configured to search the memory **156** (e.g., information related to data files stored in the memory **156**) for data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the selected product **178**. Based on the search, virtual-world server **150** may determine one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that may be used to

obtain the product **178**, and a benefit count **142/164** corresponding to each of the one or more data files that the first user **110** may receive when the data file (e.g., real-world data file **138** or virtual data file **160**) is used to obtain the product **178**.

(64) The virtual-world server **150** may be configured to indicate to the first user **110** in the virtual environment **102** (e.g., within the virtual sub-environment **120**) that the determined one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) may provide to the first user **110** the corresponding benefit counts **142/164** when the determined one or more data files are used to obtain the product **178**. For example, virtual-world server **150** may display in the virtual environment **102** (e.g. within the virtual sub-environment **120**) an identity of each of the one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the selected product **178**, and the corresponding benefit count **142/164** that may be received by the first user **110** when the data file (e.g., real-world data file **138** or virtual data file **160**) is used to obtain the selected product **178**. In one embodiment, virtual-world server **150** may additionally indicate to the first user **110** in the virtual environment **102** (e.g., virtual sub-environment **120**) an amount of real-world data objects **140** or virtual data objects **162** that are to be transferred to a corresponding target real-world data file **138** or a target virtual data file **160** respectively to obtain the selected product **178**.

(65) In certain embodiments, based on searching the information relating to data files stored in the memory **156**, virtual-world server **150** may determine multiple data files (e.g., multiple real-world data file **138**, multiple virtual data files **160** or combinations thereof) that may be used to obtain a same selected product **178**. In this case, different ones of the multiple data files **160** may provide different benefit counts **142/164** when used to obtain the same selected product **178**. The virtual-world server **150** may be configured to indicate to the first user **110** in the virtual environment **102** (e.g., within the virtual sub-environment **120**) identities of each of the multiple data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the selected product **178**, and a benefit count **142/164** corresponding to each of the data files that the first user **110** may receive when the data file is used to obtain the product **178**. In one embodiment, virtual-world server **150** may additionally indicate to the first user **110** in the virtual environment **102** (e.g., virtual sub-environment **120**) an amount of real-world data objects **140** or virtual data objects **162** that are to be transferred to a corresponding target real-world data file **138** or a target virtual data file **160** respectively to obtain the selected product **178**.

(66) In certain embodiments, when a selected product type **176** is associated with multiple products **178** that are available for selection, virtual-world server **150** may determine a single data file (e.g., real-world data file **138** or virtual data file **160**) that may be used to obtain several products **178** of the same selected product type **176** including the product **178** selected by the first user **110** of the same selected product type **176**. In this case, the single data file may provide different benefit counts **142/164** when used to obtain the different products **178** of the selected product type **176**. The virtual-world server **150** may be configured to indicate to the first user **110** in the virtual environment **102** (e.g., within the virtual sub-environment **120**) an identity of the single data file (e.g., real-world data file **138** or virtual data file **160**) that can be used to obtain each of the multiple products **178** associated with the selected product type **176**, one or more of the multiple products **178** (e.g., including the selected product **178**) associated with the selected product type **176**, and for each indicated product **178**, a benefit count **142/164** corresponding to the data file that the first user **110** may receive when the data file is used to obtain the product **178**. In one embodiment, virtual-world server **150** may additionally indicate to the first user **110** in the virtual environment **102** (e.g., virtual sub-environment **120**) an amount of real-world data objects **140** or virtual data objects **162** that are to be transferred to a corresponding target real-world data file **138** or a target virtual data file **160** respectively to obtain each of the multiple products **178**.

(67) In alternative or additional embodiments, in response to receiving a selection of a product type **176** associated with a particular selected experience theme **174**, virtual-world server **150** may be

configured to search the memory **156** and determine one or more products **178** that are available for selection associated with each selected product type **176**. For each product **178** that is available for selection, virtual-world server **150** may be configured to search the information relating to data files (e.g., real-world data file **138** and/or virtual data file **160**) stored in the memory **156** to determine one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the product **178** and a benefit count **142/164** corresponding to each determined data file that the first user **110** may receive when the data file is used to obtain the product **178**. In one embodiment, virtual-world server **150** may determine different data files (e.g., real-world data file **138** and/or virtual data file **160**) that may be used to obtain different products **178** belonging to the same product type **176**. For example, virtual-world server **150** may determine a first data file (e.g., real-world data file **138** or virtual data file **160**) that may be used to obtain a first product **178** belonging to a product type **176** and determine a second data file (e.g., real-world data file **138** or virtual data file **160**) that may be used to obtain a second product **178** belonging to the same product type **176**. The first data file may provide a first benefit count **142/164** when used to obtain the first product **178**, and the second data file may provide a second benefit count **142/164** when used to obtain the second product **178**.

(68) The virtual-world server **150** may be configured to indicate to the first user **110** in the virtual environment **102** (e.g., within the virtual sub-environment **120**) each of one or more products **178** that is available for selection associated with the selected product type **176**, identities of one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the product **178**, and a benefit count **142/164** corresponding to each of the one or more data files that the first user **110** may receive when the data file is used to obtain the product **178**. In one embodiment, virtual-world server **150** may additionally indicate to the first user **110** in the virtual environment **102** (e.g., virtual sub-environment **120**) an amount of real-world data objects **140** or virtual data objects **162** that are to be transferred to a corresponding target real-world data file **138** or a target virtual data file **160** respectively to obtain each product **178**.

(69) Virtual-world server **150** may be configured to generate a customized virtual experience for the first user **110** based on a selection of one or more products **178** made by the first user **110** in respective product types **176** of an experience theme **174**. In one embodiment, the virtual experience simulates an actual experience that the first user **110** may have when using the selected products **178** and thus allows the first user **110** to experience the selected products **178** before obtaining the products **178** by transferring data objects (e.g., real-world data objects **140** or virtual data objects **162**) to respective entities that provide the products **178**. For example, when the first user **110** selects an experience theme **174** relating to a visual experience in the virtual environment **102** as described above, the first user **110** may select several products **178** associated with multiple product types **176**. For example, the first user **110** may select a particular avatar as a selected product **178** from the “avatars” product type **176**, one or more buildings selected products **178** from the “buildings” product type **176**, and one or more roads as a selected products **178** from the “roads” product type **176**. In response to receiving these product selections, virtual-world server **150** may generate a customized virtual sub-environment **120** that simulates a visual experience of the virtual environment **102** using the selected avatar, buildings, and roads. Virtual-world server **150** may allow the first user **110** to navigate the customized virtual sub-environment **120** to experience the simulated visual experience generated for the first user **110**. In another example, when the first user **110** selects a travel theme as the selected experience theme **174**, the first user **110** may select a particular destination (e.g. city or town) as a selected product **178** from the “destination” product type **176**, a particular car as a selected product **178** from the “car” product type **176**, a particular hotel as a selected product **178** from the “hotels” product type **176**, and a particular excursion as a selected product **178** from the “excursions” product type **176**. In response to receiving these product selections associated with the travel theme, virtual-world server **150** may generate a customized virtual sub-environment **120** that simulates a travel experience of the virtual

environment **102** using the selected destination, car, hotel and excursion. Virtual-world server **150** may allow the first user **110** to navigate the customized virtual sub-environment **120** to experience the simulated travel experience generated for the first user **110**.

(70) Virtual-world server **150** may be configured to allow the first user **110** (e.g., using first avatar **114**) to select one or more data files (e.g. real-world data file **138** and/or virtual data file **160**). For example, in response to indicating to the user in the virtual environment **102** (e.g., virtual sub-environment **120**) one or more data files (e.g. real-world data file **138** and/or virtual data file **160**) that can be used to obtain one or more products **178** associated with an experience theme **174** selected by the first user **110**, first user **110** may select a data file (e.g. real-world data file **138** or virtual data file **160**). Upon receiving a selection of the data file from the first user **110**, virtual-world server **150** may be configured to initiate a registration process to register the first user **110** to obtain the selected data file. In one embodiment, when the data file selected by the first user **110** is a real-world data file **138**, the virtual-world server **150** may be configured to transmit a request to the real-world server **130** to take over and register the first user **110** to obtain the real-world data file **138**. On the other hand, when the data file selected by the first user **110** is a virtual data file **160**, the virtual-world server **150** may be configured to register the first user **110** obtain the real-world data file **138**. As described above, registering the first user **110** may include verifying the identity of the first user **110** and then generating the selected data file (e.g. real-world data file **138** or virtual data file **160**) for the first user **110**. Once a selected real-world data file **138** or a selected virtual data file **160** has been generated for the first user **110**, the first user **110** may store real-world data objects **140** or virtual data objects **162** in the respective real-world data file **138** or a selected virtual data file **160**. In one embodiment, the real-world server **130** or the virtual-world server **150** may transfer a default or pre-configured amount of real-world data objects **140** or virtual data objects **162** to the respective real-world data file **138** or a selected virtual data file **160**.

(71) Once the first user **110** is registered for the selected data file (real-world data file **138** or virtual data file **160**) and the selected data file has been generated and stored at the respective real-world server **130** or virtual-world server **150**, first user **110** may use the data file to obtain one or more products **178**. For example, when a selected product **178** is a real-world product (e.g., car, train, hotel etc.), the first user **110** may obtain the selected product **178** by commanding the real-world server **130** to transfer a pre-configured amount of real-world data objects **140** from the registered real-world data file **138** of the first user **110** to a target real-world data file **138** of an entity that provides the selected product **178**. On the other hand, when a selected product **178** is a virtual-world product (e.g., avatar, virtual building, virtual roads, etc.), the first user **110** may obtain the selected product **178** by commanding the virtual-world server **150** to transfer a pre-configured amount of virtual data objects **162** from the registered virtual data file **160** of the first user **110** to a target virtual data file **160** of an entity that provides the selected product **178** in the virtual environment **102**. In response to receiving the command from the first user **110**, the respective real-world server **130** or the virtual-world server **150** may transfer the pre-configured amount of data objects to the target data file, to cause the transfer of the selected product **178** to the first user **110** in the real-world environment or the virtual environment **102** whichever the case may be.

(72) In certain embodiment, before initiating registration of the first user **110** to obtain a selected data file (real-world data file **138** or virtual data file **160**), the virtual-world server may be configured to determine whether the first user **110** is eligible to obtain the requested data file. As described above, the virtual-world server **150** may store a user profile **172** associated with the first user **110**. Virtual-world server **150** may be configured to determine whether the first user **110** is eligible to obtain the selected data file based on the information stored in the user profile **172** associated with the first user **110**. The virtual-world server **150** initiates registration of the first user **110** to obtain the selected data file upon determining that first user **110** is eligible to obtain the selected data file.

(73) In some embodiments, the first user **110** may be registered with a first entity that owns and/or

manages the virtual environment **102** but may not be registered with a second entity that provides the data files (real-world data file **138** or virtual data file **160**). For example, based on the registration with the first entity, the first user **110** may have generated a user credential **170** that allows the first user **110** to enter the virtual environment **102**. The first user **110** may have provided some information (e.g., name, email address, social security number, phone number etc.) when registering with the first entity. This information may be stored as part of the user profile **172** of the first user **110** and associated with the user credential **170** of the first user **110**. When the first user **110** selects a data file (real-world data file **138** or virtual data file **160**) for registration, the virtual-world server **150** obtains the user profile **172** of the first user **110** associated with the user credential **170** and determines whether the first user is eligible to obtain the selected data file based on information stored in the user profile **172**.

(74) FIG. 3 illustrates a flowchart of an example method **300** for performing data interactions in a virtual environment **102**, in accordance with one or more embodiments of the present disclosure. Method **300** may be performed by the virtual-world server **150** shown in FIG. 1.

(75) At operation **302**, virtual-world server **150** receives a user credential **170** associated with a first user **110**, wherein the user credential **170** provides the first user **110** access to a virtual environment **102**.

(76) At operation **304**, virtual-world server **150** checks whether the user credential **170** is registered at the virtual-world server **150** to provide the first user **110** access to the virtual environment **102**. For example, the virtual environment **102** may be owned and/or managed by a first entity. In this case, the first user **110** may need to be registered with the first entity to gain access to the virtual environment **102**. If the first user **110** is registered with the first entity, this registration may be stored at the virtual-world server **150** including the user credential **170** that was registered with the first entity to gain access to the virtual environment **102**. If the first user **110** is found to be not authorized to access the virtual environment **102** based on the user credential **170**, method **300** proceeds to operation **306** where the virtual-world server **150** denies the first user **110** access to the virtual environment **102**. On the other hand, if the first user **110** is found to be authorized to access the virtual environment **102**, method **300** proceeds to operation **308**, where the virtual-world server **150** authorizes, an avatar (e.g., first avatar **114**) of the first user **110** to enter the virtual environment **102**.

(77) As described above, virtual-world server **150** may be configured to use the user credential **170** collected from the first user **110**, generated by the first user **110** or assigned to the first user **110** during real-world data interactions with the first user **110**, to verify identity of the first user **110** in the virtual environment **102**. Thus, the user credential **170** provides the first user **110** access to the virtual environment **102**. For example, the user credential **170** may be used by the virtual-world server **150** to verify that the first avatar **114** belongs to and is controlled by the first user **110**.

(78) In one example, user credential **170** may include a username and password generated by the first user **110** as part of registering with the real-world server **130**. The virtual-world server **150** may allow the first user **110** to use the same username and password to enter the virtual environment **102** via first avatar **114**. In an alternative embodiment, the first user **110** may generate a separate user credential (different from the first user credential **170**) as part of registering with the real-world server **130** and use the separate user credential to log into the real-world server **130** and perform real-world data interaction. The first user **110** may separately generate the user credential **170** as part of registering with the virtual-world server **150** (e.g., including registering with a first entity that owns and/or manages the virtual environment **102**), for example, during a real-world data interaction with the real-world server **130**. Once the first user **110** has registered with the virtual-world server **150** and generated the user credential **170**, the first user **110** may use the user credential **170** to log into the virtual-world server **150**, gain access to the virtual environment **102** and perform virtual-world data interactions in within the virtual environment **102**.

(79) At operation **310**, virtual-world server **150** receives a first selection made by the first user **110**

of an experience theme **174** using the avatar (e.g., first avatar **114**) in the virtual environment **102**, wherein the experience theme **174** includes a plurality of selectable product types **176**.

(80) As described above, virtual-world server **150** may be configured to facilitate a user (e.g., first user **110**) to obtain one or more products **178** in the virtual environment **102** or real-world environment. As shown in FIG. **1**, virtual-world server **150** may be configured to store one or more experience themes **174**, wherein each experience theme **174** relates to a particular type of experience that can be had by the user **110** in the virtual environment **102** or the real-world environment. An example of an experience theme associated with the virtual environment **102** may include visual elements that make up the virtual environment **102**. An example of an experience theme associated with the real-world environment may include a travel experience. An experience theme **174** includes one or more product types **176**, wherein each product type **176** is a type of product **178** that relates to the experience theme **174**. For example, when the experience theme **174** relates to visual experience of the first user **110** within the virtual environment **102**, example product types **176** associated with this experience theme **174** may include, but are not limited to, avatars, buildings, roads, objects, colors and patterns. In another example, when the experience theme **174** relates to a travel experience in the real-world environment, product types **176** associated with this experience theme **174** may include, but are not limited to, cars, buses, trains, travel destinations, hotels, restaurants, and excursions. An experience theme **174** may include several products **178** associated with each product type **176**. For example, when the experience theme **174** relates to a visual experience within the virtual environment **102**, the “avatars” product type **176** may include several designs of avatars **114** that may be used as the first avatar **114** of the first user **110** within the virtual environment **102**. The “buildings” product type **176** may include a plurality of virtual buildings that can be used to generate the virtual environment **102**. The “colors” and “patterns” product type **176** may include a plurality of colors and patterns respectively that can be used to render specific objects or portions of the virtual environment **102**. When the experience theme **174** relates to a travel experience in the real-world environment, the “destination” product type **176** may include a selection of different destinations (e.g., cities, towns etc.) the first user **110** may visit, the “cars” product type **176** may include a selection of different cars that be obtained for a travel, the “hotels” product type may include a selection of hotels at the selected destination and the like.

(81) When the first user **110** enters the virtual environment **102** (e.g., using first avatar **114**) after logging in to the virtual-world server **150** by providing user credential **170**, the first user **110** may navigate to a virtual sub-environment **120** that is configured to facilitate users to obtain products **178** associated with one or more experience themes **174**. The virtual sub-environment **120** may be assigned to a particular real-world entity. Once the first user **110** has entered the virtual sub-environment **120**, the virtual-world server **150** may present to the first user **110** one or more experience themes **174**. For example, the virtual-world server **150** may display the experience themes **174** on a virtual display within the virtual sub-environment **120**. Virtual-world server **150** may be configured to allow the first user **110** to select one of the presented experience themes **174**. The first user **110** may select one of the experience themes **174** depending on what kind of experience the first user **110** is looking for. For example, the virtual-world server **150** may present a visual experience theme associated with the virtual environment **102** and a travel experience theme associated with the real-world environment. In an example, the first user **110** selects the visual experience theme associated with the virtual environment **102**.

(82) At operation **312**, virtual-world server **150** receives a second selection made by the first user **110** in the virtual environment **102** of at least one product type **176** from the selected experience theme **174**, wherein a first product **178** associated with the selected at least one product type **176** can be obtained by the first user **110** by transferring a first amount of data objects (e.g., real-world data objects **140** and/or virtual data objects **162**) to a first entity that provides the first product **178**.

(83) As described above, once the virtual-world server **150** receives a selection of an experience

theme **174**, virtual-world server **150** may be configured to present to the first user **110** within the virtual sub-environment **120**, one or more product types **176** associated with the selected experience theme **174**. For example, when the first user **110** selects visual experience in the virtual environment **102** as the selected experience theme **174**, virtual-world server **150** presents a list of product types **176** associated with the visual experience theme including, but not limited to, avatars, buildings, roads, objects, colors and patterns. Virtual-world server **150** may be configured to allow the first user **110** to select one or more of the presented product types **176** associated with the selected experience theme **174**. As the first user **110** selects each product type **176**, virtual-world server **150** may be configured to access from the memory **156** one or more products **178** associated with the selected product type **176** and present the one or more products **178** to the first user **110** in the virtual environment **102**. For example, when the first user **110** selects the “avatars” product type **176**, virtual-world server **150** may access from the memory **156** one or more avatar designs associated with the “avatars” product type **176** and present the one or more avatar designs to the first user **110** in the virtual-sub environment **120**. For example, virtual-world server **150** may display a 2-dimensional or 3-dimensional image of one or more avatars associated with the avatar product type **176**. Virtual-world server **150** may allow the first user **110** to select at least one product **178** (e.g., specific avatar design) associated with a selected product type **176**. In one embodiment, virtual-world server **150** may allow the first user **110** to select multiple product types **176** related to the selected experience theme **174** and one or more products **178** associated with each of the multiple selected product types **176**. For example, the first user **110** may want to build a customized virtual sub-environment **120** and may want to obtain specific visual elements (e.g., avatars, buildings, roads, objects, colors, patterns etc.) to build the virtual sub-environment **120**. For this purpose, the first user **110** may select a specific avatar from the “avatar” product type **176**, one or more specific buildings from the “buildings” product type **176**, one or more roads from the “roads” product type **176**, one or more colors from the “colors” product type **176**, and one or more patterns from the “patterns” product type **176**.

(84) At operation **314**, virtual-world server **150** determines a first data file (e.g., real-world data file **138** or virtual data file **160**) that provides a first benefit count (e.g., benefit count **142/164**) to the first user **110** when the first amount of data objects (e.g., real-world data objects **140** or virtual data objects **162**) is transferred from the first data file to a second data file (e.g., another real-world data file **138** or virtual data file **160**) associated with the first entity to obtain the first product **178** associated with the selected at least one product type **176**.

(85) As described above, once the product selections have been made by the first user **110**, virtual-world server **150** may be configured to determine one or more data files (e.g., real-world data files **138** and/or virtual data files **160**) that can be used to obtain at least one of the products **178** selected by the first user **110**. In this context, virtual-world server **150** may be configured to store information relating to a plurality of real-world data files **138** and virtual data files **160** that may be used by the first user **110** to obtain products **178** associated with experience themes **174**. A particular product **178** may be obtained by the first user **110** by transferring a pre-configured amount of real-world data objects **140** from a real-world data file **138** registered to the first user **110** to another real-world data file **138** registered to an entity that provides the particular product **178**, and/or by transferring a pre-configured amount of virtual data objects **162** from a virtual data file **160** registered to the first user **110** to another virtual data file **160** registered to the entity that provides the particular product **178**. Different products **178** may be obtained by transferring different pre-configured amounts of real-world data objects **140** or virtual data objects **162** to respective data files of entities that provide the products **178**. For example, a first product **178** may be obtained by the first user **110** by transferring a first pre-configured amount of real-world data objects **140** or virtual data objects **162**, and a second product **178** may be obtained by the first user **110** by transferring a second pre-configured amount of real-world data objects **140** or virtual data objects **162**.

(86) In some embodiments, a single data file (e.g., real-world data file **138** or virtual data file **160**) may be used to obtain multiple products **178**. In some embodiments, any one of multiple data files (e.g., multiple real-world data file **138**, multiple virtual data files **160** or combinations thereof) may be used to obtain a same product **178**.

(87) In one or more embodiments, a data file (e.g., real-world data file **138** or virtual data file **160**) may provide a benefit count **142/164** to the first user **110**, when the first user **110** uses the data file (e.g., real-world data file **138** or virtual data file **160**) to obtain a particular product **178** (e.g., by transferring data objects from the data file to a respective data file of an entity that provides the particular product **178**). For example, the first user **110** may receive a pre-configured benefit count **142** when the first user **110** uses a particular real-world data file **138** to obtain a particular product **178**. Similarly, the first user **110** may receive a pre-configured benefit count **164** when the first user **110** uses a particular virtual data file **160** to obtain a particular product **178**. In one embodiment, different data files (e.g., different real-world data files **138** and/or different virtual data files **160**) may provide different benefit counts **142/164** to obtain a same product **178**. In an additional or alternative embodiment, a single data file (e.g., real-world data file **138** or virtual data file **160**) may provide different benefit counts **132/164** to obtain different products **178**.

(88) In one or more embodiments, benefit count **142** associated with a real-world data file **138** may be used to obtain one or more products **178** in the real-world environment, and benefit count **164** associated with a virtual data file **160** may be used to obtain one or more products **178** in the virtual environment **102**. A higher benefit count **142/164** generally allows the first user **110** to obtain more products **178**. Thus, a higher benefit count **142/164** is generally of higher value. The first user **110** may accumulate benefit counts **142/164** received as a result of using a particular data file (e.g., real-world data file **138** or virtual data file **160**) to obtain several products **178**.

(89) The information stored by the virtual-world server **150** in memory **156** relating to a plurality of real-world data files **138** and virtual data files **160** that may be used to obtain products **178** associated with experience themes **174** may include one or more of an identity of each real-world data file **138** or virtual data file **160**, products **178** that may be obtained using the real-world data file **138** or the virtual data file **160**, and benefit counts **142** or **164** provided by the real-world data file **138** or virtual data file **160** respectively when used to obtain each of the products **178**.

(90) As described above, once the selection of the products **178** associated with a selected experience theme **174** has been made by the first user **110**, virtual-world server **150** may be configured to determine one or more data files (e.g., real-world data files **138** or virtual data files **160**) that can be used to obtain one or more of the products **178** selected by the first user **110**. For a selected product **178**, virtual-world server **150** determines one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that may be used to obtain the product **178**, and a benefit count **142/164** corresponding to each of the one or more data files that the first user **110** may receive when the data file (e.g., real-world data file **138** or virtual data file **160**) is used to obtain the product **178**. For example, based on the selected product **178**, virtual-world server **150** may be configured to search the memory **156** (e.g., information related to data files stored in the memory **156**) for data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the selected product **178**. Based on the search, virtual-world server **150** may determine one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that may be used to obtain the product **178**, and a benefit count **142/164** corresponding to each of the one or more data files that the first user **110** may receive when the data file (e.g., real-world data file **138** or virtual data file **160**) is used to obtain the product **178**.

(91) At operation **316**, virtual-world server **150** indicates to the first user **110** in the virtual environment **102** that the first data file (e.g., real-world data file **138** or virtual data file **160**) provides the first benefit count **142/164** to the first user **110** in relation to obtaining the first product **178** associated with the selected at least one product type **176**, wherein the indication comprises the first benefit count **142/164**.

(92) The virtual-world server **150** may be configured to indicate to the first user **110** in the virtual environment **102** (e.g., within the virtual sub-environment **120**) that the determined one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) may provide to the first user **110** the corresponding benefit counts **142/164** when the determined one or more data files are used to obtain the product **178**. For example, virtual-world server **150** may display in the virtual environment **102** (e.g. within the virtual sub-environment **120**) an identity of each of the one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the selected product **178**, and the corresponding benefit count **142/164** that may be received by the first user **110** when the data file (e.g., real-world data file **138** or virtual data file **160**) is used to obtain the selected product **178**. In one embodiment, virtual-world server **150** may additionally indicate to the first user **110** in the virtual environment **102** (e.g., virtual sub-environment **120**) an amount of real-world data objects **140** or virtual data objects **162** that are to be transferred to a corresponding target real-world data file **138** or a target virtual data file **160** respectively to obtain the selected product **178**.

(93) In certain embodiments, based on searching the information relating to data files stored in the memory **156**, virtual-world server **150** may determine multiple data files (e.g., multiple real-world data file **138**, multiple virtual data files **160** or combinations thereof) that may be used to obtain a same selected product **178**. In this case, different ones of the multiple data files **160** may provide different benefit counts **142/164** when used to obtain the same selected product **178**. The virtual-world server **150** may be configured to indicate to the first user **110** in the virtual environment **102** (e.g., within the virtual sub-environment **120**) identities of each of the multiple data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the selected product **178**, and a benefit count **142/164** corresponding to each of the data files that the first user **110** may receive when the data file is used to obtain the product **178**. In one embodiment, virtual-world server **150** may additionally indicate to the first user **110** in the virtual environment **102** (e.g., virtual sub-environment **120**) an amount of real-world data objects **140** or virtual data objects **162** that are to be transferred to a corresponding target real-world data file **138** or a target virtual data file **160** respectively to obtain the selected product **178**.

(94) In certain embodiments, when a selected product type **176** is associated with multiple products **178** that are available for selection, virtual-world server **150** may determine a single data file (e.g., real-world data file **138** or virtual data file **160**) that may be used to obtain several products **178** of the same selected product type **176** including the product **178** selected by the first user **110** of the same selected product type **176**. In this case, the single data file may provide different benefit counts **142/164** when used to obtain the different products **178** of the selected product type **176**. The virtual-world server **150** may be configured to indicate to the first user **110** in the virtual environment **102** (e.g., within the virtual sub-environment **120**) an identity of the single data file (e.g., real-world data file **138** or virtual data file **160**) that can be used to obtain each of the multiple products **178** associated with the selected product type **176**, one or more of the multiple products **178** (e.g., including the selected product **178**) associated with the selected product type **176**, and for each indicated product **178**, a benefit count **142/164** corresponding to the data file that the first user **110** may receive when the data file is used to obtain the product **178**. In one embodiment, virtual-world server **150** may additionally indicate to the first user **110** in the virtual environment **102** (e.g., virtual sub-environment **120**) an amount of real-world data objects **140** or virtual data objects **162** that are to be transferred to a corresponding target real-world data file **138** or a target virtual data file **160** respectively to obtain each of the multiple products **178**.

(95) In alternative or additional embodiments, in response to receiving a selection of a product type **176** associated with a particular selected experience theme **174**, virtual-world server **150** may be configured to search the memory **156** and determine one or more products **178** that are available for selection associated with each selected product type **176**. For each product **178** that is available for selection, virtual-world server **150** may be configured to search the information relating to data

files (e.g., real-world data file **138** and/or virtual data file **160**) stored in the memory **156** to determine one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the product **178** and a benefit count **142/164** corresponding to each determined data file that the first user **110** may receive when the data file is used to obtain the product **178**. In one embodiment, virtual-world server **150** may determine different data files (e.g., real-world data file **138** and/or virtual data file **160**) that may be used to obtain different products **178** belonging to the same product type **176**. For example, virtual-world server **150** may determine a first data file (e.g., real-world data file **138** or virtual data file **160**) that may be used to obtain a first product **178** belonging to a product type **176** and determine a second data file (e.g., real-world data file **138** or virtual data file **160**) that may be used to obtain a second product **178** belonging to the same product type **176**. The first data file may provide a first benefit count **142/164** when used to obtain the first product **178**, and the second data file may provide a second benefit count **142/164** when used to obtain the second product **178**.

(96) The virtual-world server **150** may be configured to indicate to the first user **110** in the virtual environment **102** (e.g., within the virtual sub-environment **120**) each of one or more products **178** that is available for selection associated with the selected product type **176**, identities of one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain the product **178**, and a benefit count **142/164** corresponding to each of the one or more data files that the first user **110** may receive when the data file is used to obtain the product **178**. In one embodiment, virtual-world server **150** may additionally indicate to the first user **110** in the virtual environment **102** (e.g., virtual sub-environment **120**) an amount of real-world data objects **140** or virtual data objects **162** that are to be transferred to a corresponding target real-world data file **138** or a target virtual data file **160** respectively to obtain each product **178**.

(97) At operation **318**, virtual-world server **150** receives a command from the first user **110** to register the first user **110** to obtain the first data file (e.g., real-world data file **138** or virtual data file **160**).

(98) At operation **320**, in response to receiving the command, virtual-world server **150** initiates a registration process to register the first user **110** to obtain the first data file (e.g., real-world data file **138** or virtual data file **160**).

(99) As described above, virtual-world environment **150** may be configured to allow the first user **110** (e.g., using first avatar **114**) to select one or more data files (e.g. real-world data file **138** and/or virtual data file **160**). For example, in response to indicating to the user in the virtual environment **102** (e.g., virtual sub-environment **120**) one or more data files (e.g. real-world data file **138** and/or virtual data file **160**) that can be used to obtain one or more products **178** associated with an experience theme **174** selected by the first user **110**, first user **110** may select a data file (e.g. real-world data file **138** or virtual data file **160**). Upon receiving a selection of the data file from the first user **110**, virtual-world server **150** may be configured to initiate a registration process to register the first user **110** to obtain the selected data file. In one embodiment, when the data file selected by the first user **110** is a real-world data file **138**, the virtual-world server **150** may be configured to transmit a request to the real-world server **130** to take over and register the first user **110** to obtain the real-world data file **138**. On the other hand, when the data file selected by the first user **110** is a virtual data file **160**, the virtual-world server **150** may be configured to register the first user **110** to obtain the real-world data file **138**. As described above, registering the first user **110** may include verifying the identity of the first user **110** and then generating the selected data file (e.g. real-world data file **138** or virtual data file **160**) for the first user **110**. Once a selected real-world data file **138** or a selected virtual data file **160** has been generated for the first user **110**, the first user **110** may store real-world data objects **140** or virtual data objects **162** in the respective real-world data file **138** or a selected virtual data file **160**. In one embodiment, the real-world server **130** or the virtual-world server **150** may transfer a default or pre-configured amount of real-world data objects **140** or virtual data objects **162** to the respective real-world data file **138** or a selected virtual data file **160**.

(100) Once the first user **110** is registered for the selected data file (real-world data file **138** or virtual data file **160**) and the selected data file has been generated and stored at the respective real-world server **130** or virtual-world server **150**, first user **110** may use the data file to obtain one or more products **178**. For example, when a selected product **178** is a real-world product (e.g., car, train, hotel etc.), the first user **110** may obtain the selected product **178** by commanding the real-world server **130** to transfer a pre-configured amount of real-world data objects **140** from the registered real-world data file **138** of the first user **110** to a target real-world data file **138** of an entity that provides the selected product **178**. On the other hand, when a selected product **178** is a virtual-world product (e.g., avatar, virtual building, virtual roads, etc.), the first user **110** may obtain the selected product **178** by commanding the virtual-world server **150** to transfer a pre-configured amount of virtual data objects **162** from the registered virtual data file **160** of the first user **110** to a target virtual data file **160** of an entity that provides the selected product **178** in the virtual environment **102**. In response to receiving the command from the first user **110**, the respective real-world server **130** or the virtual-world server **150** may transfer the pre-configured amount of data objects to the target data file, to cause the transfer of the selected product **178** to the first user **110** in the real-world environment or the virtual environment **102** whichever the case may be.

(101) In certain embodiment, before initiating registration of the first user **110** to obtain a selected data file (real-world data file **138** or virtual data file **160**), the virtual-world server may be configured to determine whether the first user **110** is eligible to obtain the requested data file. As described above, the virtual-world server **150** may store a user profile **172** associated with the first user **110**. Virtual-world server **150** may be configured to determine whether the first user **110** is eligible to obtain the selected data file based on the information stored in the user profile **172** associated with the first user **110**. The virtual-world server **150** initiates registration of the first user **110** to obtain the selected data file upon determining that first user **110** is eligible to obtain the selected data file.

(102) In some embodiments, the first user **110** may be registered with a first entity that owns and/or manages the virtual environment **102** but may not be registered with a second entity that provides the data files (real-world data file **138** or virtual data file **160**). For example, based on the registration with the first entity, the first user **110** may have generated a user credential **170** that allows the first user **110** to enter the virtual environment **102**. The first user **110** may have provided some information (e.g., name, email address, social security number, phone number etc.) when registering with the first entity. This information may be stored as part of the user profile **172** of the first user **110** and associated with the user credential **170** of the first user **110**. When the first user **110** selects a data file (real-world data file **138** or virtual data file **160**) for registration, the virtual-world server **150** obtains the user profile **172** of the first user **110** associated with the user credential **170** and determines whether the first user is eligible to obtain the selected data file based on information stored in the user profile **172**.

(103) In an example banking use case, the disclosed system and methods described in embodiments of the present disclosure provide motivations to a user (e.g., first user **110**) to sign up for a bank account (e.g., a credit card). In this context, the real-world data file **138** may correspond to a real-world bank account of the first user **110** and the real-world data objects **140** may correspond to the real-world funds in the bank account of the first user **110**. Similarly, virtual data file **160** may correspond to a virtual account (e.g., a crypto account) of the first user **110** and the virtual data objects **162** may correspond to digital currency. Different types of real-world data file **138** may include, but are not limited to, credit card account, debit card account, savings account, and checking account. Benefit count **164** may correspond to reward points that may be earned by the first user **110** when a particular bank account (e.g., credit card or crypto account) is used to purchase a product.

(104) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific

forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

(105) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(106) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system comprising: a memory that stores a user credential associated with a first user; and at least one processor coupled to the memory, and configured to: receive the user credential associated with the first user, wherein the user credential provides the first user access to a virtual environment; authorize, based on the user credential, an avatar of the first user to enter the virtual environment; receive a first selection made by the first user of an experience theme using the avatar in the virtual environment, wherein the experience theme comprises a plurality of selectable product types; receive a second selection made by the first user in the virtual environment of at least one product type from the selected experience theme, wherein a first product associated with the selected at least one product type can be obtained by the first user by transferring a first amount of data objects to a first entity that provides the first product; determine a first data file that provides a first benefit count to the first user when the first amount of data objects is transferred from the first data file to a second data file associated with the first entity to obtain the first product associated with the selected at least one product type; indicate to the first user in the virtual environment that the first data file provides the first benefit count to the first user in relation to obtaining the first product associated with the selected at least one product type, wherein the indication comprises the first benefit count; receive a command from the first user to register to obtain the first data file; and in response to receiving the command, initiate a registration process to register the first user to obtain the first data file.

2. The system of claim 1, wherein the at least one processor is further configured to: after the user has obtained the first data file, receive a second command from the first user to transfer the first amount of data objects from the first data file to the second data file associated with the entity to obtain the first product; and in response to receiving the second command, transfer the first amount of data objects from the first data file to the second data file, to cause a transfer of the first product to the first user in a real-world environment.

3. The system of claim 2, wherein the at least one processor is further configured to: determine whether the first user is registered with a second entity associated with the virtual environment; if the first user is not registered with the second entity, obtain a user profile of the first user associated with the user credential of the first user; determine, based on the user profile, whether the first user is eligible to obtain the first data file; and initiate a registration process to register the first user to obtain the first data file, if the first user is eligible to obtain the first data file.

4. The system of claim 1, wherein the at least one processor is further configured to: generate a

virtual sub-environment within the virtual environment, wherein the virtual sub-environment provides the first user a virtual experience associated with one or more products associated with the selected at least one product type.

5. The system of claim 1, wherein: the selected experience theme comprises a plurality of products associated with the selected at least one product type; a second product associated with the selected at least one product type can be obtained by the first user by transferring a second amount of data objects to a second entity that provides the second product; and the at least one processor is further configured to: determine a third data file that provides a second benefit count to the first user when the second amount of data objects is transferred from the third data file to a fourth data file of the second entity to obtain the second product associated with the selected at least one product type, wherein the first benefit count is larger than the second benefit count; and indicate to the first user in the virtual environment that the third data file provides the second benefit count to the first user in relation to obtaining the second product associated with the selected at least one product type, wherein the indication comprises the second benefit count.

6. The system of claim 5, wherein the at least one processor is further configured to: determine that the first data file provides a third benefit count to the first user when the second number of data objects is transferred from the first data file to the fourth data file of the second entity to obtain the second product associated with the selected at least one product type, wherein the first benefit count is larger than the third benefit count; and indicate to the first user in the virtual environment that the first data file provides the third benefit count to the first user in relation to obtaining the second product associated with the selected at least one product type, wherein the indication comprises the third benefit count.

7. The system of claim 1, wherein the at least one processor is further configured to: determine that a third data file provides a second benefit count to the first user when the first amount of data objects is transferred from the third data file to the second data file associated with the first entity to obtain the first product associated with the selected at least one product type, wherein the first benefit count is larger than the second benefit count; and indicate to the first user in the virtual environment that the third data file provides the second benefit count to the first user in relation to obtaining the first product associated with the selected at least one product type, wherein the indication comprises the second benefit count.

8. The system of claim 1, wherein: the second selection of the at least one product type comprises: a third selection of a first product type; and a fourth selection of a second product type; and the at least one processor is further configured to: generate a virtual sub-environment within the virtual environment, wherein the virtual sub-environment provides the first user a virtual experience associated with at least one product associated with the first product type and at least one product associated with the second product type.

9. A method for performing data interactions in a virtual environment, the method comprising: receiving a user credential associated with the first user, wherein the user credential provides the first user access to a virtual environment; authorizing, based on the user credential, an avatar of the first user to enter the virtual environment; receiving a first selection made by the first user of an experience theme using the avatar in the virtual environment, wherein the experience theme comprises a plurality of selectable product types; receiving a second selection made by the first user in the virtual environment of at least one product type from the selected experience theme, wherein a first product associated with the selected at least one product type can be obtained by the first user by transferring a first amount of data objects to a first entity that provides the first product; determining a first data file that provides a first benefit count to the first user when the first amount of data objects is transferred from the first data file to a second data file associated with the first entity to obtain the first product associated with the selected at least one product type; indicating to the first user in the virtual environment that the first data file provides the first benefit count to the first user in relation to obtaining the first product associated with the selected at least

one product type, wherein the indication comprises the first benefit count; receiving a command from the first user to register to obtain the first data file; and in response to receiving the command, initiating a registration process to register the first user to obtain the first data file.

10. The method of claim 9, further comprising: after the user has obtained the first data file, receiving a second command from the first user to transfer the first amount of data objects from the first data file to the second data file associated with the entity to obtain the first product; and in response to receiving the second command, transferring the first amount of data objects from the first data file to the second data file, to cause a transfer of the first product to the first user in a real-world environment.

11. The method of claim 10, further comprising: determining whether the first user is registered with a second entity associated with the virtual environment; if the first user is not registered with the second entity, obtaining a user profile of the first user associated with the user credential of the first user; determining, based on the user profile, whether the first user is eligible to obtain the first data file; and initiating a registration process to register the first user to obtain the first data file, if the first user is eligible to obtain the first data file.

12. The method of claim 9, further comprising: generating a virtual sub-environment within the virtual environment, wherein the virtual sub-environment provides the first user a virtual experience associated with one or more products associated with the selected at least one product type.

13. The method of claim 9, wherein: the selected experience theme comprises a plurality of products associated with the selected at least one product type; a second product associated with the selected at least one product type can be obtained by the first user by transferring a second amount of data objects to a second entity that provides the second product; and further comprising: determining a third data file that provides a second benefit count to the first user when the second amount of data objects is transferred from the third data file to a fourth data file of the second entity to obtain the second product associated with the selected at least one product type, wherein the first benefit count is larger than the second benefit count; and indicating to the first user in the virtual environment that the third data file provides the second benefit count to the first user in relation to obtaining the second product associated with the selected at least one product type, wherein the indication comprises the second benefit count.

14. The method of claim 13, further comprising: determining that the first data file provides a third benefit count to the first user when the second number of data objects is transferred from the first data file to the fourth data file of the second entity to obtain the second product associated with the selected at least one product type, wherein the first benefit count is larger than the third benefit count; and indicating to the first user in the virtual environment that the first data file provides the third benefit count to the first user in relation to obtaining the second product associated with the selected at least one product type, wherein the indication comprises the third benefit count.

15. The method of claim 9, further comprising: determining that a third data file provides a second benefit count to the first user when the first amount of data objects is transferred from the third data file to the second data file associated with the first entity to obtain the first product associated with the selected at least one product type, wherein the first benefit count is larger than the second benefit count; and indicating to the first user in the virtual environment that the third data file provides the second benefit count to the first user in relation to obtaining the first product associated with the selected at least one product type, wherein the indication comprises the second benefit count.

16. The method of claim 9, wherein: the second selection of the at least one product type comprises: a third selection of a first product type; and a fourth selection of a second product type; and further comprising: generating a virtual sub-environment within the virtual environment, wherein the virtual sub-environment provides the first user a virtual experience associated with at least one product associated with the first product type and at least one product associated with the

second product type.

17. A non-transitory computer-readable medium that stores instructions which when executed by a processor causes the processor to: receive a user credential associated with the first user, wherein the user credential provides the first user access to a virtual environment; authorize, based on the user credential, an avatar of the first user to enter the virtual environment; receive a first selection made by the first user of an experience theme using the avatar in the virtual environment, wherein the experience theme comprises a plurality of selectable product types; receive a second selection made by the first user in the virtual environment of at least one product type from the selected experience theme, wherein a first product associated with the selected at least one product type can be obtained by the first user by transferring a first amount of data objects to a first entity that provides the first product; determine a first data file that provides a first benefit count to the first user when the first amount of data objects is transferred from the first data file to a second data file associated with the first entity to obtain the first product associated with the selected at least one product type; indicate to the first user in the virtual environment that the first data file provides the first benefit count to the first user in relation to obtaining the first product associated with the selected at least one product type, wherein the indication comprises the first benefit count; receive a command from the first user to register to obtain the first data file; and in response to receiving the command, initiate a registration process to register the first user to obtain the first data file.

18. The non-transitory computer-readable medium of claim 17, wherein the instructions further cause the processor to: after the user has obtained the first data file, receive a second command from the first user to transfer the first amount of data objects from the first data file to the second data file associated with the entity to obtain the first product; and in response to receiving the second command, transfer the first amount of data objects from the first data file to the second data file, to cause a transfer of the first product to the first user in a real-world environment.

19. The non-transitory computer-readable medium of claim 18, wherein the instructions further cause the processor to: determine whether the first user is registered with a second entity associated with the virtual environment; if the first user is not registered with the second entity, obtain a user profile of the first user associated with the user credential of the first user; determine, based on the user profile, whether the first user is eligible to obtain the first data file; and initiate a registration process to register the first user to obtain the first data file, if the first user is eligible to obtain the first data file.

20. The non-transitory computer-readable medium of claim 17, wherein the instructions further cause the processor to: generate a virtual sub-environment within the virtual environment, wherein the virtual sub-environment provides the first user a virtual experience associated with one or more products associated with the selected at least one product type.
